FINTECH PRODUCT & ARCHITECTURE DECK

# FinWise AI

A production-grade, AI-powered personal financial planning platform combining deterministic calculation engines with transparent multi-agent financial reasoning.

- ⚡ Next.js 14 App Router
- 🐍 FastAPI & SQLAlchemy
- 🤖 Multi-Agent Orchestration
- ₹ INR & 50/30/20 Rules
- 🛡️ Human-in-the-Loop AI

# Why Personal Finance Remains Broken

Millions of young professionals and households struggle with financial fragmentation, lack of predictive planning, and opaque financial advice.



Problem 01

## Passive Spreadsheets

Traditional expense trackers only show where money already vanished. They lack forward-looking simulation, cash-flow stress testing, and real-time guidance.



Problem 02

## Opaque & Risky Advice

Black-box generic chatbots hallucinate numbers and give risky financial suggestions without verifying real balances, debts, or essential liquidity constraints.



Problem 03

## Fragmented Portfolios

Users juggle separate apps for budgeting, EMI tracking, and goal planning, making it impossible to answer: *"Can I truly afford this ₹50,000 purchase today?"*

# The Three Pillars of FinWise AI

FinWise bridges human decision-making and artificial intelligence with three tightly coupled core systems.



Pillar 1

## PFM Dashboard

High-clarity financial command center tailored for Indian consumers (INR ₹). Real-time surplus calculation, net cash flow, categorical budgets, and liquidity scores.

- ✓ Automated 50/30/20 categorization
- ✓ Live cash flow & monthly trends
- ✓ Visual progress gauges on goals



Pillar 2

## Deterministic Engine

Exact, verifiable financial logic written in pure Python with decimal precision. Zero hallucination for formulas, debt schedules, and scenario projections.

- ✓ Avalanche vs Snowball debt payoff
- ✓ Compounding SIP growth modeling
- ✓ Cash runway & shock stress-tests



Pillar 3

## Multi-Agent AI Copilot

Collaborative agent swarm analyzing authorized financial records. Generates transparent step-by-step reasoning logs and requires approval for sensitive actions.

- ✓ 10 financial reasoning clusters
- ✓ Full execution audit trail
- ✓ Human-in-the-loop approval cards

# Comprehensive Financial Management

A unified platform with 22 dedicated routes designed for full financial lifecycle management.



### Transactions

Smart ledger with CSV batch import, manual entry modals, type filters (Income, Expense, Transfer), and export capabilities.



### Savings Goals

Emergency funds, vacation pools, and tech purchases with target dates, auto-calculated monthly contributions, and progress tracking.



### Debt Planner

Interactive simulator comparing mathematically optimal **Avalanche** (highest APR) vs psychologically rewarding **Snowball**.



### What-If Forecasts

Monte Carlo-inspired simulation modeling income increases, job transitions, rent changes, and one-off expense impacts over 12 months.



### Monthly Reports

Executive summaries, category pie breakdowns, and printable PDF financial statements for easy review and tax filing.



### Recurring Bills

Subscription tracking, auto-reminders, and due-date calendars ensuring zero late-payment penalties or forgotten trials.



### Security & Privacy

Granular data consent, one-click JSON data export, and full GDPR/DPDP-compliant account deletion workflows.



### System Admin

Role-based administrator console for system health metrics, active user management, and agent execution trace auditing.

# Inside the Multi-Agent Financial Assistant

A domain-specific multi-agent architecture delivering institutional-grade financial guidance with zero hallucination.

## 10 Specialized Financial Reasoning Clusters

Every query is semantically parsed and assigned to specialized agents with live database context:

1. Purchase Affordability
2. Spending & Outflow Auditing
3. 50/30/20 Budget Optimization
4. Emergency Buffer Sizing
5. Debt Avalanche / Snowball
6. Wealth Building & SIPs
7. Indian Tax Regime Planning
8. Cashflow Shocks & Runway
9. Inflation Purchasing Power
10. Behavioral Financial Coaching

## Safety & Explainability by Design

Unlike raw LLM prompts, FinWise enforces strict guardrails before presenting any guidance to the user:

### 1. Real Portfolio Verification

Injects verified SQL metrics (Net Income: ₹60,500, Surplus: ₹28,050, Liquid Savings: ₹57,700). The AI cannot invent hypothetical balance figures.

### 2. Transparent Agent Execution Traces

Every response displays the exact agents invoked, data sources queried, and decision-making logic in real time.

### 3. Human Approval Requirement

Any automated recommendation (e.g. creating a budget cap or moving funds) requires explicit user approval before database modification.

# Modern, Scalable Technology Stack

Constructed with high-performance frameworks supporting strict type safety, modular architecture, and rapid cloud deployment.

## Frontend Architecture

Built with Next.js 14 and React 18 for lightning-fast server-rendered performance and accessibility.

- ✓ `Next.js 14` App Router architecture
- ✓ `TypeScript` with strict type verification
- ✓ `Tailwind CSS` with custom fintech palette
- ✓ `Recharts` interactive financial charts
- ✓ `Lucide React` accessible iconography

## Backend Architecture

Robust asynchronous Python service layer with validation, migrations, and automated testing.

- ✓ `FastAPI` high-throughput ASGI framework
- ✓ `Pydantic v2` request & response schemas
- ✓ `SQLAlchemy 2.0` ORM with connection pooling
- ✓ `Alembic` repeatable database migrations
- ✓ `Pytest` automated backend test suites

## Database & Infrastructure

Financial-grade database design with explicit timezone and numeric precision guarantees.

- ✓ `MySQL 8 / SQLite` ACID compliance
- ✓ `DECIMAL(14,2)` precision currency handling
- ✓ `Vercel Edge` global CDN deployment
- ✓ `Render / Docker` containerized backend
- ✓ `Cloudflare` SSL reverse proxy tunneling

# Enterprise Security & Data Protection

Architected following defense-in-depth security principles to safeguard sensitive financial information.

## Authentication & Identity

No custom or unproven encryption algorithms. All authentication flows follow modern industry standards:

- ✓ **Bcrypt Password Hashing:** Salted, computationally intensive hashing resistant to rainbow table and brute-force attacks.
- ✓ **Stateless JWT Tokens:** Signed with 256-bit secret keys and strict 7-day expiration windows.
- ✓ **Role-Based Access Control (RBAC):** Strict separation between standard end-users and administrators.
- ✓ **Secure Password Recovery:** Cryptographic reset tokens with single-use verification.

## Network & Transport Security

Zero plain-text transmissions and robust API boundary protection:

- ✓ **Unified Reverse Proxy:** Frontend proxies all `/api/v1/*` calls internally, eliminating CORS exposure and mixed-content SSL blocks.
- ✓ **Server-Side Authorization:** Every backend endpoint independently validates user identity and entity ownership.
- ✓ **Tamper-Resistant Audit Logging:** Sensitive financial events and agent actions are persisted in an immutable audit ledger.
- ✓ **User Data Privacy:** Complete user autonomy with instant data export and GDPR/DPDP-compliant account deletion.

# Tested & Verified Live Execution

Demonstration data verified with production test workflows and benchmarked financial indicators.

**₹60,500**
Monthly Net Income

**₹15,879**
Monthly Expenses

**₹44,621**
Net Monthly Cash Flow

**73.8%**
Healthy Savings Rate

## Pre-configured Demo Accounts

| Role | Name | Email | Password |
|---|---|---|---|
| User | John Smith | `demo@finwise.ai` | `Demo1234!` |
| Admin | System Admin | `admin@finwise.ai` | `Admin1234!` |

## Tested User Workflows

- ✓ **Affordability Query:** AI analyzed laptop purchase against ₹44,621 surplus and approved purchase with 2-month buffer rule.
- ✓ **Debt Payoff Comparison:** Avalanche simulated ₹14,200 total interest savings vs standard minimum payments.
- ✓ **Automated Seeding:** Instant demonstration database generation with zero manual configuration.

# Product Roadmap & Expansion

Strategic evolution from an intelligent planning platform to a fully automated personal finance operating system.

Phase 1 (Completed)

## Core PFM & AI Engine

- ✓ Full 22 Next.js App Router routes
- ✓ FastAPI deterministic calculation engine
- ✓ 10-cluster semantic AI reasoning coordinator
- ✓ Vercel & Cloudflare production hosting

Phase 2 (Next 90 Days)

## Account Aggregator & Sync

- ✓ RBI-licensed Account Aggregator integration
- ✓ Automated bank statement & UPI sync
- ✓ Real-time mutual fund & stock portfolio tracking
- ✓ WhatsApp financial alert bot & digest

Phase 3 (Long-term)

## Global Expansion & Voice

- ✓ Multi-currency engine (USD, EUR, GBP, AED)
- ✓ Gemini Live bidirectional voice financial copilot
- ✓ Automated tax filing integration (ITR-1 & ITR-2)
- ✓ Collaborative household & couple budgeting

READY FOR PRODUCTION

# Make Smarter Decisions With Your Money.

FinWise AI is fully implemented, end-to-end tested, and accessible today with production code, automated seeding, and multi-agent financial reasoning.

**LIVE DEPLOYMENT**
**https://finwise-ai-demo.vercel.app**

**DEMO CREDENTIALS**
demo@finwise.ai / Demo1234!

🚀 Ready to Scale

📑 Complete Documentation

💻 100% Full-Stack Source Code